



County Level MIPS Quality Scores: Coverage Limits and Association with Medicare Outcomes

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Introduction

Established by Centers for Medicare & Medicaid Services, the Merit-based Incentive Payment System (MIPS) is a system designed to transition Medicare clinicians towards a value-based care system. Given the strong financial incentives present for clinicians to best mimic the behavior MIPS rewards, it is in Medicare users’ benefit for as strong a correlation between high MIPS scores and positive patient outcomes as possible. We aim to explore whether MIPS is financially incentivizing physician behavior that creates better short- and long-term outcomes for patients.

Methods

We downloaded three separate data files to determine if MIPS predicts various patient outcomes. During data collection, we evaluated three files for missingness, structure, and column names to ensure reliable merging. We identified physician groups with multiple ZIP codes and confirmed the files contained the necessary data for group-level performance and regional outcome analyses, including inverse performance measures. During data sanitization, we prepared the newly combined data by standardizing column names and ZIP codes, then used sequential left joins to combine the MIPS, DAC, and patient outcomes data into the final DataFrame.

For unsupervised learning, we restricted the dataset to 2023 values only and re-shaped it to one row per county and one column per MIPS measure. Measures reported in fewer than 30% of counties and counties reporting fewer than 20% of remaining measures were removed. Missing values were median imputed, and measures were standardized. Principal Component Analysis (PCA) retained the minimum components explaining at least 90% of cumulative variance. Silhouette scores for K-means clustering with k = 2 through 10 were used to select the optimal k, after which clusters were compared on Medicare geographic outcomes using Kruskal Wallis tests.

For supervised learning, we used the same restricted dataset from unsupervised learning and applied the same 30% measure and 20% county coverage thresholds, yielding 2,507 counties and 49 measures. We applied leakage guards for circular measures, sibling strata, and identity features, removing six measures, and sign adjusted six inverse measures at display time. The MIPS increment was defined as the difference in cross-validated r^2 between models with county controls alone and controls plus MIPS measures. We trained LinearRegression, ElasticNetCV, RandomForestRegressor, and GradientBoostingRegressor models using GroupKFold with “state” as the grouping variable and compared model, feature, and target combinations to select the optimal approach.

Results

Of 229 MIPS measures, 55 met the 30% county coverage threshold, and 1,400 of 2,955 counties met the 20% coverage threshold for clustering. PCA required 41 components to explain at least 90% of variance. K-means clustering produced the highest silhouette score at k = 2 (0.096), yielding imbalanced clusters of 445 and 955 counties. Despite weak cluster separation, 28 of 44 Medicare geographic outcomes differed at $p < 0.05$ by Kruskal Wallis tests, with the strongest differences in utilization and cost measures and weaker separation for acute outcomes such as readmission and emergency room utilization.

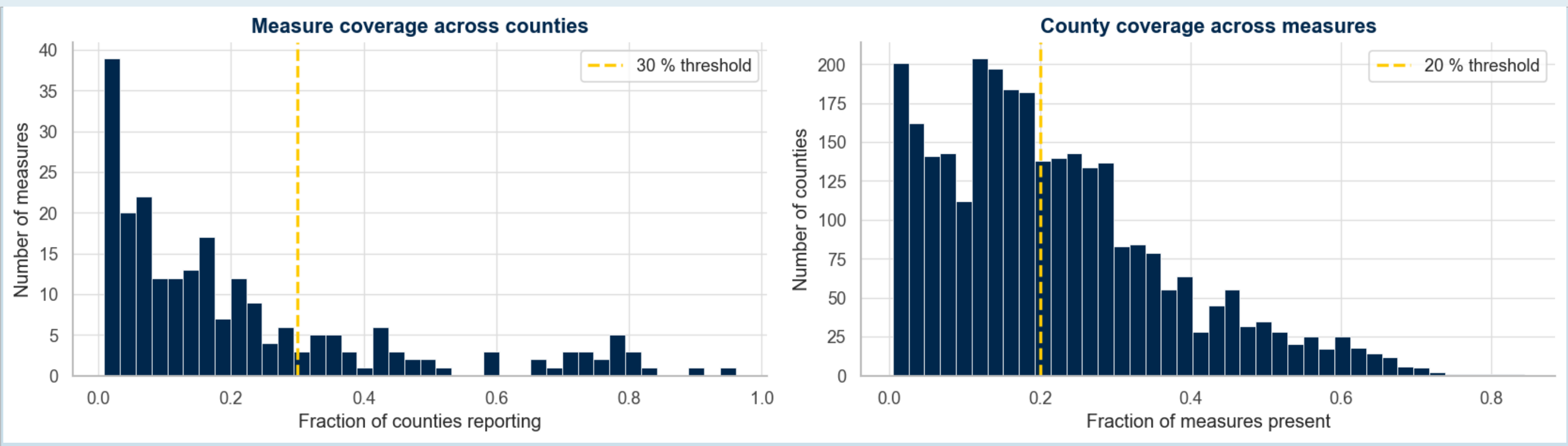


Figure 1 Measure coverage across counties and county coverage across measures.

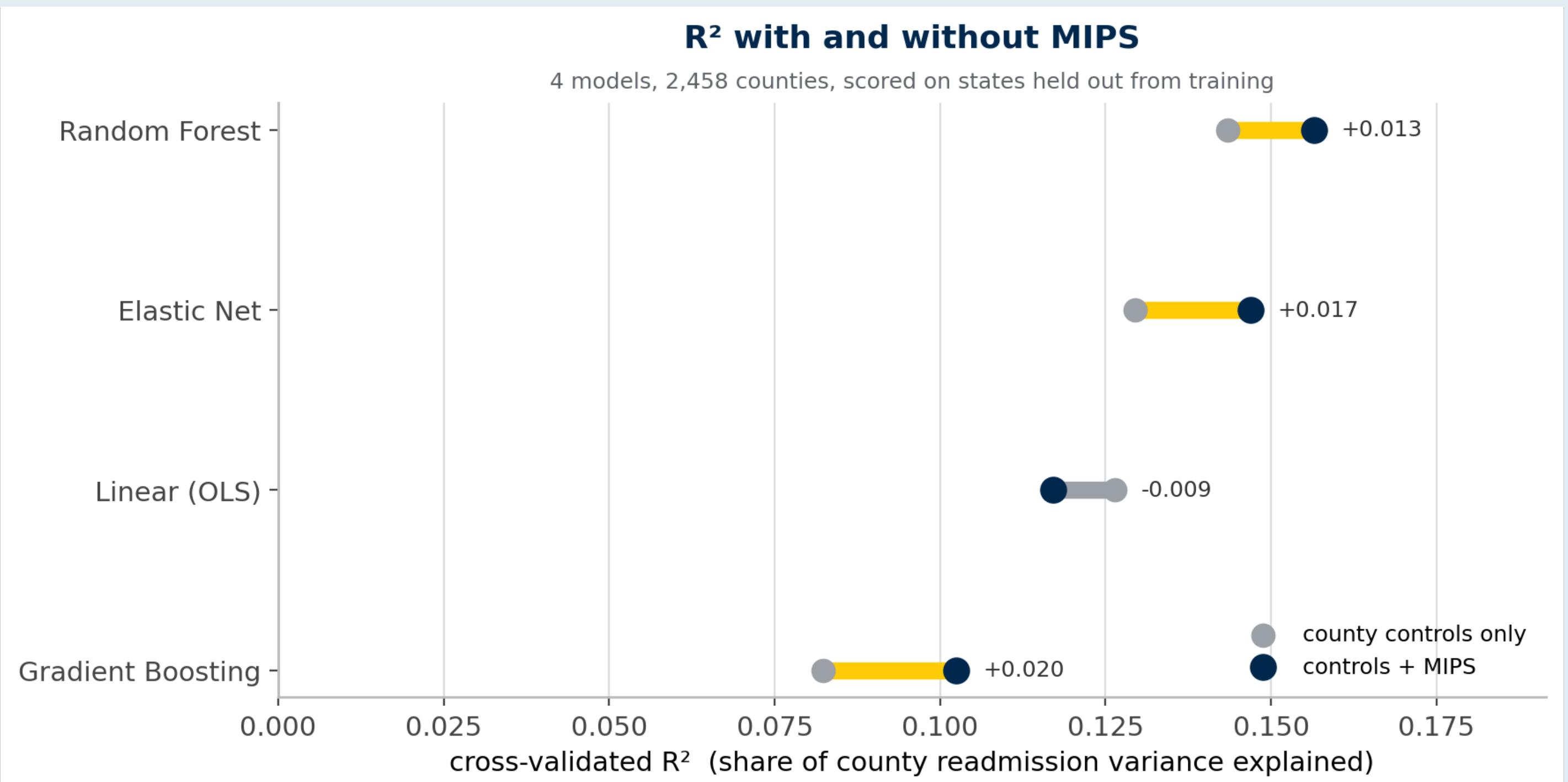


Figure 2 R^2 values with and without MIPS.

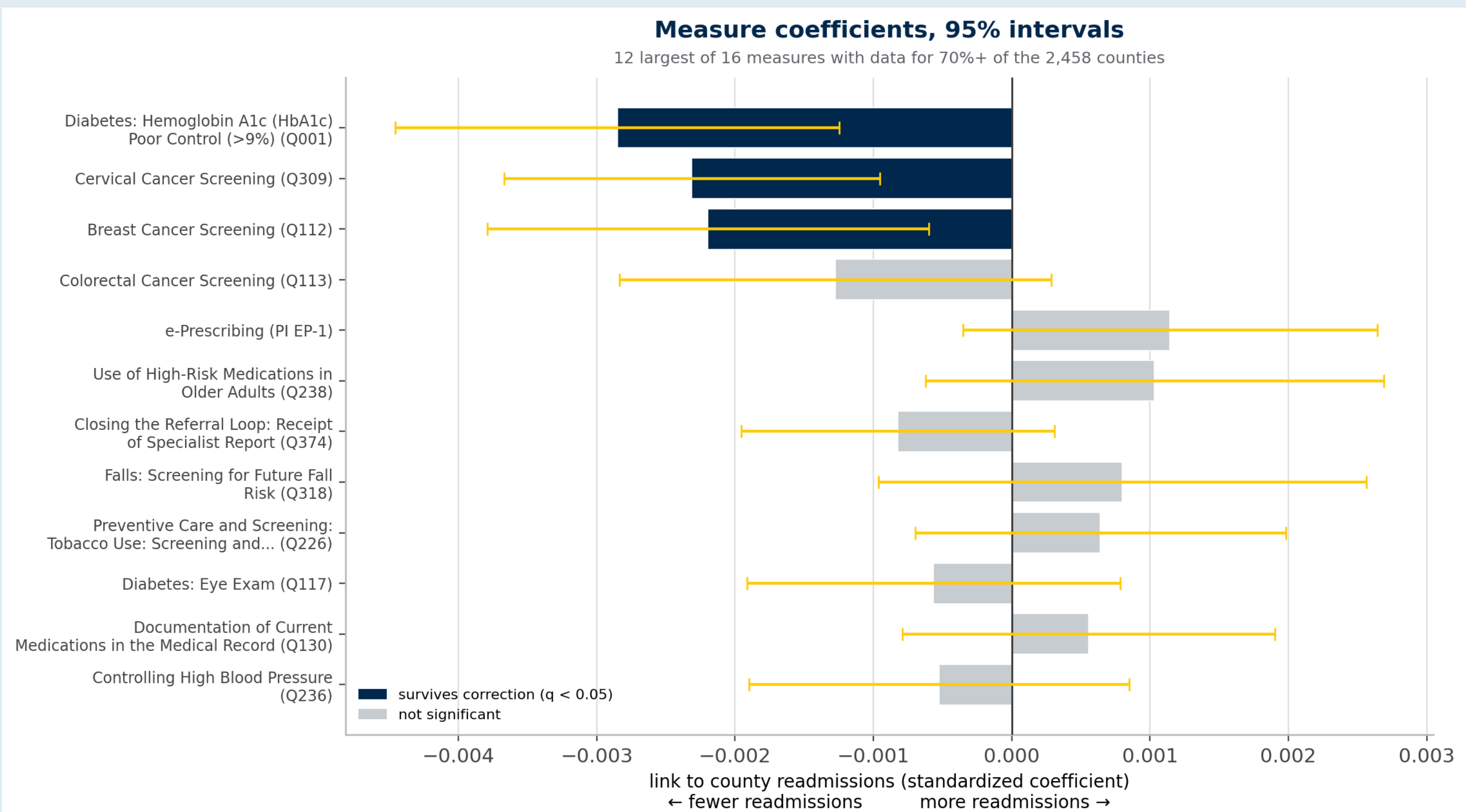


Figure 3 The 12 largest of the 16 measures clearing the 70% coverage floor, with 95% intervals. Blue indicates the three measures that survive BH correction.

Results (cont.)

We evaluated two targets, four models, and two feature sets, producing 18 combinations. Elastic Net yielded a MIPS increment of $+0.017$ r^2 (95% CI [-0.000, +0.035], $p = .050$), with positive gains across all five held-out folds. Gradient boosting showed the largest increment, but reporting indicators recovered 81.6% of that gain, suggesting that reporting patterns may account for much of the added signal in the model. Using 2023 MIPS data for adjacent-year outcomes produced increments of $+0.043$ (2022), $+0.020$ (2023), and $+0.019$ (2024), suggesting persistent county characteristics rather than a MIPS effect. Prior-year readmission rate reduced the MIPS increment from $+0.006$ to $+0.002$ r^2 when both were included. With a 70% coverage threshold, three measures remained significant after Benjamini-Hochberg correction: HbA1c poor control, cervical cancer screening, and breast cancer screening, all associated with fewer readmissions. Residuals showed spatial autocorrelation (Moran’s I = $+0.199$, $p = .001$); all three remained significant with Conley spatial HAC errors, while one remained significant under census-division clustering.

Discussion

PCA required 41 components to explain 90% of variance across the 55 retained MIPS measures, indicating that county performance varies across many dimensions rather than a few dominant components. The best K-means solution (k = 2) had a low silhouette score (0.096) and imbalanced clusters (445 and 955 counties), suggesting a multidimensional continuum rather than clearly separated MIPS performance groups. Despite weak separation, 28 of 44 Medicare geographic outcomes differed at unadjusted $p < 0.05$, with the strongest differences in utilization and cost measures and weaker differences in acute outcomes such as readmission and emergency room utilization. These findings suggest MIPS performance is more closely associated with practice patterns and resource utilization than acute clinical events. Interpretation is limited by exclusion of approximately 53% of counties for insufficient MIPS coverage, median imputation of approximately 20% of retained values, and the observational, cross-sectional design, which limits causal inference.

County-aggregated MIPS showed only a weak relationship with Medicare outcomes, and spares reporting limits its use as a regional quality signal. 52.0% of county measure cells come from a single reporting practice and 69.2% from two or fewer, so scores often reflect limited participation rather than county-wide performance. The stronger association with outcomes preceding the performance year, along with spatially clustered residuals, further suggests stable county characteristics and regional factors rather than a MIPS effect. These county-level associations do not reflect individual practice performance, practices are weighted by ZIP-to-county allocation rather than patient volume, and coverage filtering may limit representativeness.

Future Work

Although our findings did not suggest a strong correlation between MIPS and patient outcomes, future work could test different county and measure cutoffs and conduct more granular analyses, as county-level comparisons may obscure differences in population density. Another opportunity is comparing MIPS with the prior Medicare reimbursement system; applying our analysis to older reimbursement data while adjusting for inflation and other time-based factors could assess whether MIPS provides greater financial incentives for physicians.